Oliver Krauss

Basics of Machine Learning

AIN2IL | MTD.ba SS2026

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FH OBERÖSTERREICH UNIVERSITY OF APPLIED SCIENCES UPPER AUSTRIA

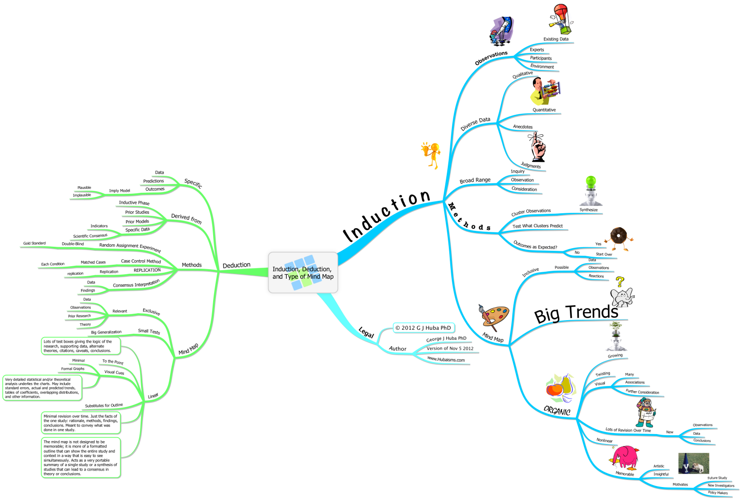
How do we solve these tasks?

- Finding solutions of a system of equations
- Trajectory prediction of a space shuttle
- Diagnosis whether a patient has a certain disease
- Prediction of election outcomes
- Recognition of handwritten characters
- Identification of customer target groups
- Prediction of protein functions from its amino acid sequence

Explicit Models

- Traditional disciplines like physics, chemistry, and biology are usually aiming at *exact explicit models*, i.e. to know how (and why) things work in a particular way; then a solution to a new problem can be found *deductively* using explicit knowledge.
- That goal, however, is sometimes too difficult to achieve; reasons may be computational complexity, insufficient knowledge, insufficient information, etc.

Inductive vs Deductive Reasoning



Source: George J Huba (Hubaisms.com)

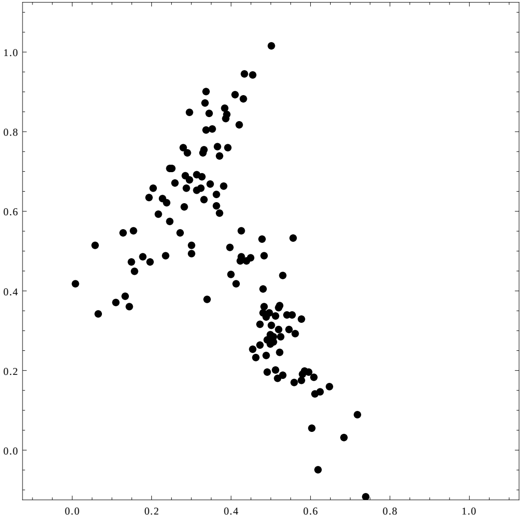
Machine Learning == Inductive Learning

- Machine learning tries to elicit models/knowledge from *previously observed data* with the following two main goals:
  1. Getting insight
  2. Being able to predict future outcomes
- Putting it simple, machine learning is about *learning from data* (often called *inductive learning*).

What Do We See Here?

0.843475	0.709216	-1
0.408987	0.47037	+1
0.734759	0.645298	-1
0.972187	0.0802574	+1
0.90267	0.327633	-1
0.807075	0.872155	-1
0.240068	0.801159	-1
0.206602	0.562109	+1
0.581611	0.335561	+1
0.700995	0.517267	-1
0.209818	0.342484	+1
0.94141	0.928017	-1
0.148546	0.198177	+1
0.872544	0.50608	-1
0.371062	0.272064	+1
...	...	...

And Here?



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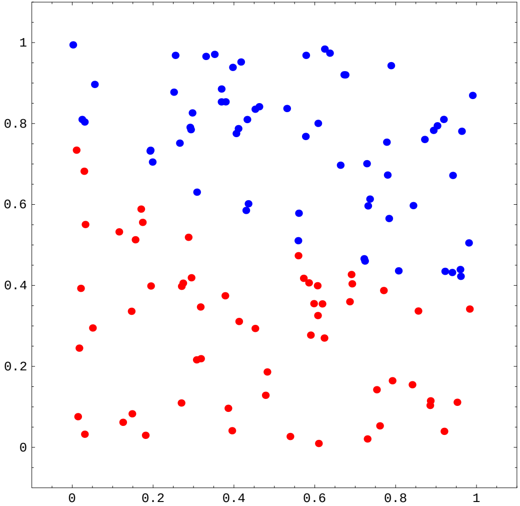

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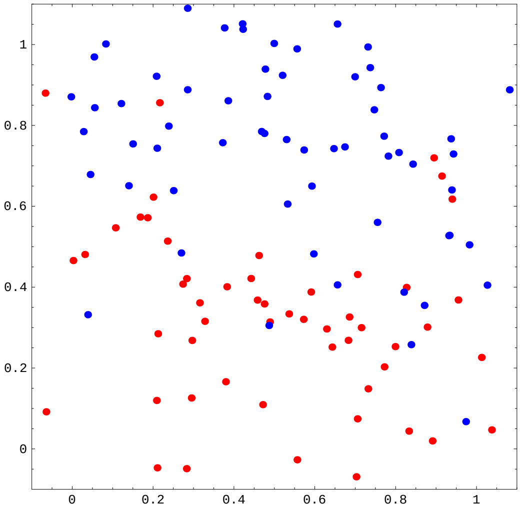

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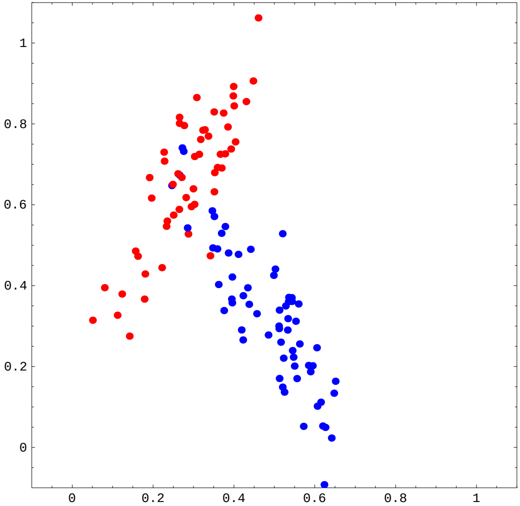
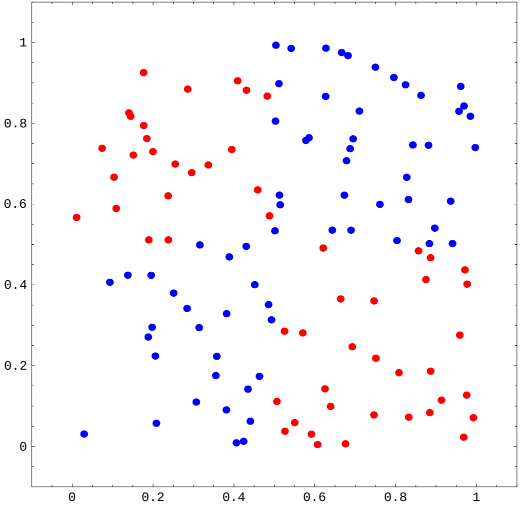


And Here?



And Here?

0.99516	0.890813	0.933726	0.793397	0.826405	0.236946	-1
0.853206	0.611647	0.317486	0.633609	0.411492	0.985231	+1
0.387494	0.459847	0.815049	0.394526	0.678227	0.031886	-1
0.733515	0.640438	1.19068	0.639685	0.0793674	0.160503	+1
0.274817	0.261054	1.20056	0.689895	0.401913	0.277955	-1
0.329943	0.241299	0.848705	0.721673	0.973852	0.795238	-1
0.334784	0.350487	0.315131	0.928277	0.816343	0.558292	-1
0.481578	0.738839	0.0925513	0.294667	0.612725	0.573062	-1
0.0940846	0.278992	0.451819	0.900141	0.220497	0.541176	+1
0.421025	0.785714	0.449038	0.920612	0.420418	0.749187	-1
0.939446	0.0468747	0.15846	0.625944	0.198894	0.176125	+1
0.845362	0.767883	0.824993	0.725803	0.808218	0.63495	-1
0.484793	0.129329	0.0783719	0.465347	0.291457	0.254278	+1
0.399041	0.751829	0.763511	0.894785	0.47902	0.15156	-1
0.643232	0.615629	0.430261	0.0458972	0.446513	0.844081	+1
...	...	...	...	...	...	...

Example 1: Empirical Determination of Concentration

- How can we measure the unknown **concentration of an aqueous solution**?
- Lambert-Beer's law states that the quantity of light absorbed by a substance dissolved in a fully transmitting solvent is directly proportional to the concentration of the substance and the path length of the light through the solution.
- **Solution:**
  1. Prepare samples of known concentrations
  2. Measure absorption using a spectrophotometer
  3. Determine proportionality coefficient from measured data
  4. Infer unknown concentration(s) from measurements

Example 1: Empirical Determination of Concentration

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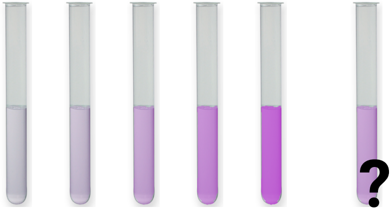

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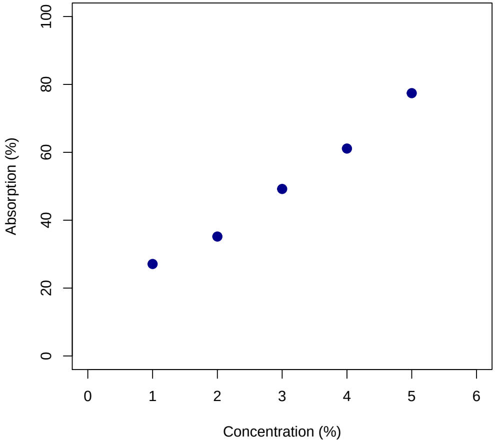

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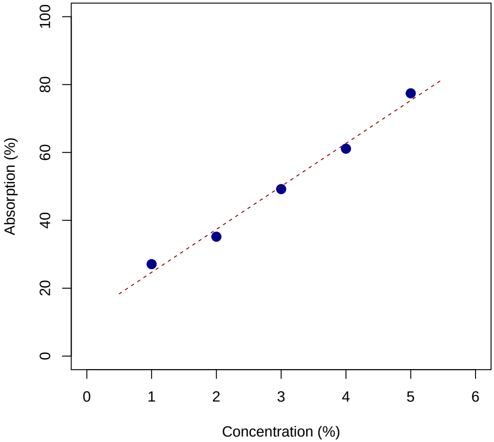

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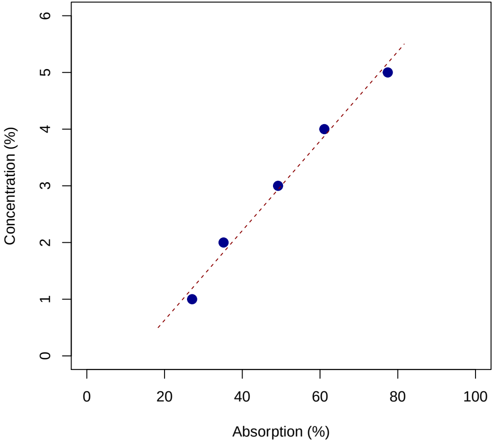

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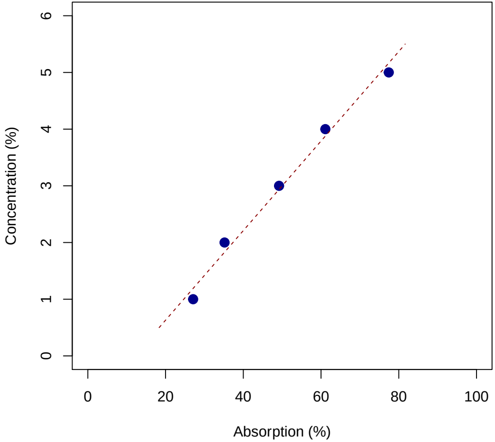

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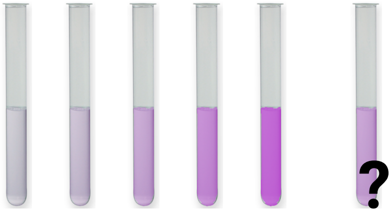
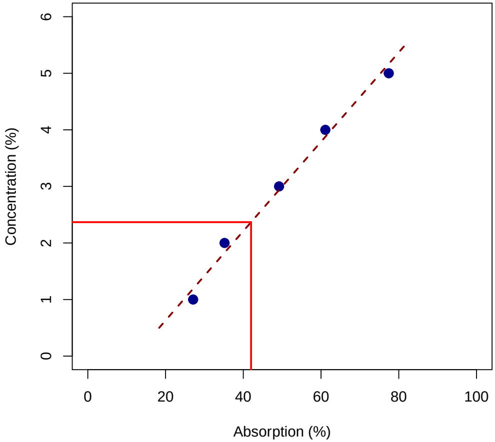




Example 2: How much is the Fish?

- Example borrowed from

  R. O. Duda, P. E. Hart, and D. G. Stork. Pattern Classification. *2nd edition. John Wiley & Sons, 2001. ISBN 0-471-05669-3.*

- Automated system to **sort fish** in a fish-packing company: salmons must be distinguished from sea bass optically
- Given: a set of **pictures with known fish**, the training set
- Goal: automatically **distinguish between** salmons and sea bass for future pictures

Our Fish

Salmon:

Figure: Source: Sydney Fish Market

Sea bass:

Figure: Source: CATTEL GmbH

Our Fish

Salmon:

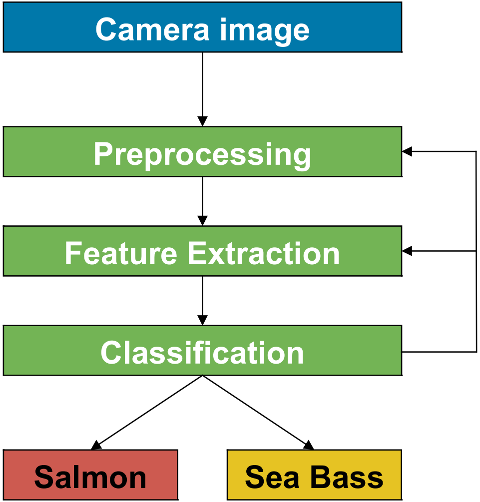
Figure: Source: Sydney Fish Market

Sea bass:

Figure: Source: CATTEL GmbH

How can we distinguish these two kinds of fish visually?

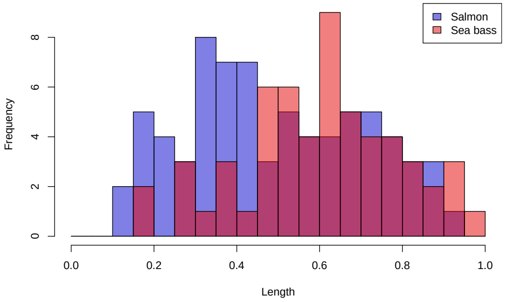
Basic Workflow

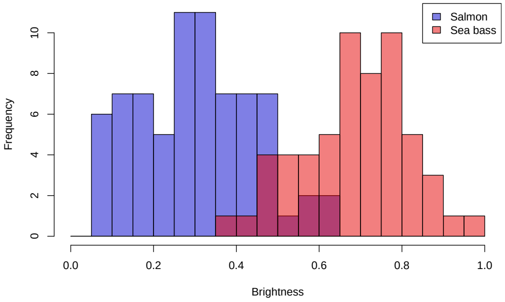


Preprocessing: contrast and brightness correction, segmentation, alignment

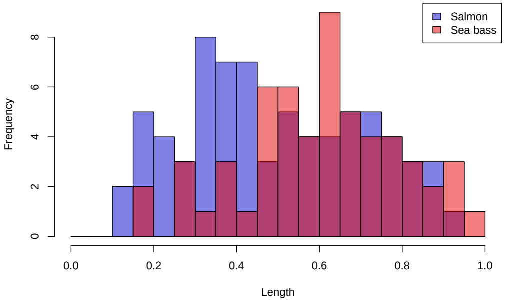
Features:
1. Length
2. Brightness

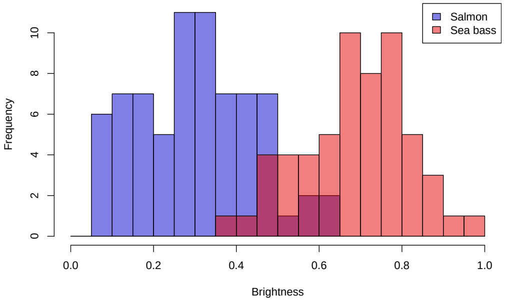
Using one Feature





Using one Feature

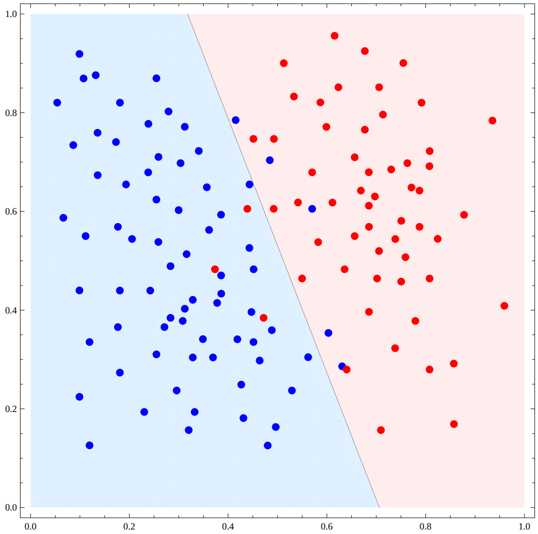




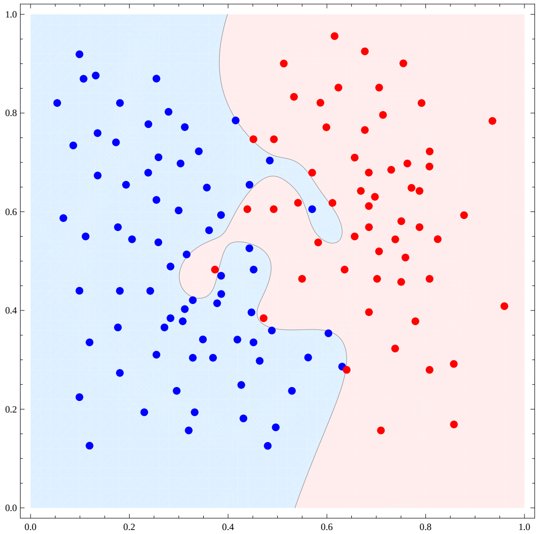
Questions:

1. Which is the **better feature**?
2. Where should we put the **threshold**?

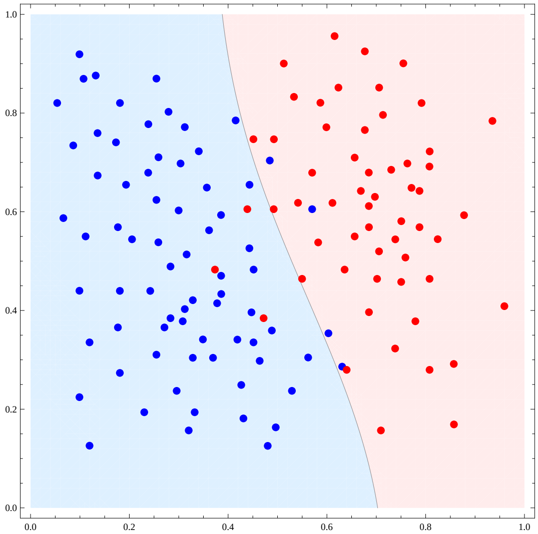
Two Features: Linear Separation



Two Features: Highly Nonlinear Separation



Two Features: Mildly Nonlinear Separation



Questions

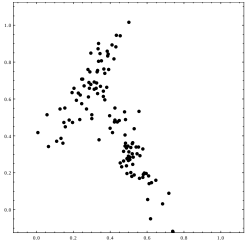
- Does **learning** help in the **future**? I.e. does experience from previously observed examples help us solve a future task?
- What is a **good model**?
- How do we asses the **quality** of a model?
- Which **methods** are available?

Machine Learning is not (only) about describing previously observed data, but about *creating models that are applicable to future data.*

Supervised vs. Unsupervised Machine Learning

Unsupervised Machinelearning:

The goal ist to **identify structure** in the data. There is no explicit target/label given.



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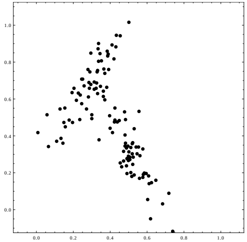
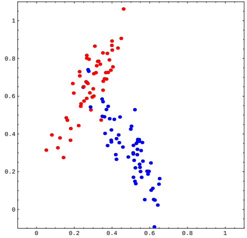


Supervised Machinelearning:

An explicit target/label value is given for each input data item. The goal is to **identify the relationship** between inputs and targets.



Unsupervised Machine Learning

Projection Methods down-projection of data to lower-dimensional space in order to concentrate on the essence of the data

Clustering grouping of similar data items

Biclustering simultaneous grouping of samples and features

Generative model building a model that produces data that is distributed the same as the observed data

Supervised Machine Learning

Classification the target value is a class label

Regression the target value is numerical

Supervised ML is sometimes called *predictive modelling*, because the goal is most often to predict the target value for future input values.

Miscellaneous Topics

Reinforcement Learning learning by feeback from the environment in an online process

Feature Extraction computation of features from data prior to machine learning (e.g. signal / image processing)

Feature Selection selection of those features that are relevant and sufficient to solve a given learning task

Feature construction construction of new features as part of the learning process

Terminology

Model the specific relationship or representation we are aming at

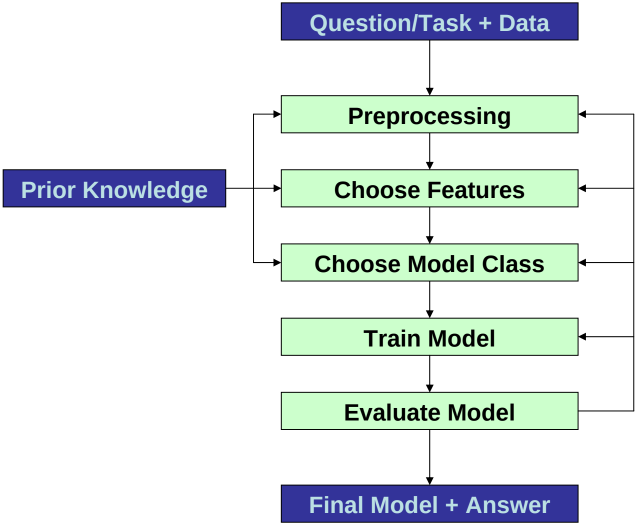
Model Class the class of models in which we search for the model

Parameters representations of concrete models inside the given model class

Model Selection/Training process of finding that model from the model class that fits/explains the observed data in the best way

Hyperparameters parameters controlling the model complexity or the training procedure

Basic Data Analysis Workflow



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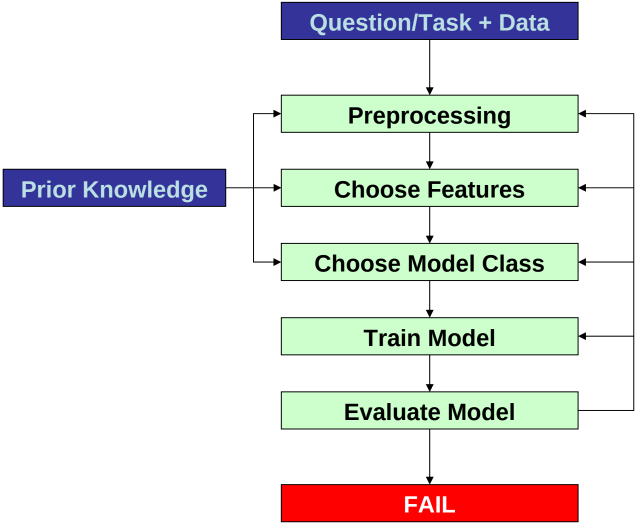

Cross-Industry Standard Process for Data Mining (CRISP-DM)

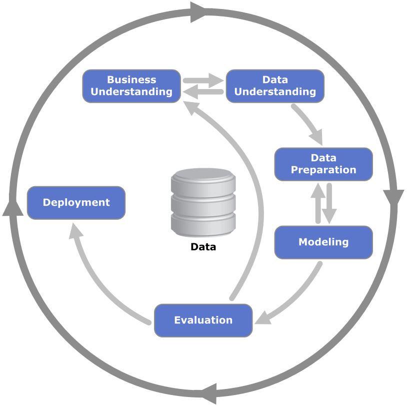

Figure: Source: Kenneth Jensen, CC BY-SA 3.0

Model Selection/Training: Basic Ingredients

For both supervised and unsupervised machine learning, we need the following basic ingredients:

Model class the class of models in which we search for the model

Objective criterion/measure that determines what a good model is

Optimization algorithm method that tries to find model parameters such that the objective is optimized

The right choices of the above components depends on the characteristics of the given task.

Words of Enthusiasm

- Machine learning methods are able to solve some **tasks for which explicit models will never exist**.
- Machine learning methods have become **standard tools** in a variety of disciplines (e.g. signal and image processing, bioinformatics).

Words of Caution

- Machine learning is **not a universal remedy**.
- The quality of a machine learning model depends on the **quality and quantity of data**.
- What **cannot be measured/observed** can never be identified by machine learning.
- Machine learning **complements explicit/deductive models** instead of replacing them.
- Machine learning is often applied in a **naive** way.